4. CV of the researcher

Surajit Bera

Contact Info and Personal Details

Position Postdoctoral Fellow

Email surajit.bera@college-de-france.fr, iamsb1995@gmail.com

Phone (+91) 9980616076

Address 11 Pl. Marcelin Berthelot, 75231 Paris, France

Date of Birth 30th October, 1995

Nationality Indian

Experience

- **Centre Nationale de la Recherche Scientifique (CNRS) - College de France** Paris, France
  Post-doctoral Fellow 01/05/2024-present
  —– Young Physics Teams of College de France, Theory of Non-Equilibrium Quantum Matter group led by Prof. Marco Schiro

- **Indian Institute of Science (IISc)** Bengaluru, India
  Doctoral Fellow and Teaching Assistance 30/08/2018-06/02/2024
  —–Department Physical Science, Condensed matter theory group, Prof. Sumilan Banerjee.

Education

- **Indian Institute of Science(IISc)** Bengaluru, India
  Integrated Ph.D (M.Sc degree+Ph.D) in Physics. 30/08/2016-06/02/2024
  Thesis for Ph.D: *Chaos, entanglement, and non-equilibrium dynamics in Hermitian and non-Hermitian quantum many-body systems*, under supervision of Prof. Sumilan Banerjee. Date of successful PhD defence: 06/02/2024.
  Passed 2 years of Master degree (Res.) in physics with CGPA: 8.6

- **Ramakrishna Mission Vivekananda Centenary College** Kolkata, India
  Bachelor degree in Science (B.Sc) (Major Physics) 12/07/2013-24/06/2016
  Passed with First division scoring 82% (Equivalent CGPA: 8.2)

Publications

1. Shashank Kumar Ojha, Sankalpa Hazra, **Surajit Bera**, Sanat Kumar Gogoi, Prithwijit Mandal, Jyotirmay Maity, A. Gloskovskii, C. Schlueter, Smarajit Karmakar, Manish Jain, Sumilan Banerjee, Venkatraman Gopalan, and Srimanta Middey, *"Quantum fluctuations lead to glassy electron dynamics in the good metal regime of electron doped KTaO3"*. **Nature Communications, 15, 3830 (2024)**

2. **Surajit Bera**, Arijit Haldar, and Sumilan Banerjee, *"Dynamical mean-field theory for Rényi entanglement entropy and mutual information in the Hubbard model"*. **Phys. Rev. B 109, 035156 (2024)**

3. **Surajit Bera**, K.Y. Venkata Lokesh, and Sumilan Banerjee, *"Quantum-to-Classical Crossover in Many-Body Chaos and Scrambling from Relaxation in a Glass"*, **Phys. Rev. Lett.** 128, 115302 (2022)

4. Arijit Haldar, **Surajit Bera**, and Sumilan Banerjee, *"Rényi entanglement entropy of Fermi and non-Fermi liquids: Sachdev-Ye-Kitaev model and dynamical mean field theories"*, **Phys. Rev. Research** 2, 033505 (2020)

5. Arijit Haldar, Prosenjit Haldar, **Surajit Bera**, Ipsita Mandal, and Sumilan Banerjee, *"Quench, thermalization, and residual entropy across a non-Fermi liquid to Fermi liquid transition"* **Phys. Rev. Research** 2, 013307 (2020)

Computational Skills

Programming Python, Matlab, Mathematica, Fortran, C

International Conferences, Schools, Visits and Presentation

- **Quantum Dynamics: From Electrons to Qbits** — Italy
  International Centre for theoretical Physics (**ICTP**) — 22/08/2022-09/09/2022
  Poster Presentation: *Renyi entanglement entropy in Hubbard model within dynamical mean field theory*

- **Visit to Institute of Science and Technology** (**ISTA**) — Austria
  Interaction and discussion with Prof. Maksym Serbyn. — 12/09/2022-14/09/2022
  Discussed *Quantum-to-Classical Crossover in Many-Body Chaos and Information Scrambling* & *New Path Integral Formulation of Entanglement Entropy*, and interacted with his group members.

- **Bangalore School on Statistical Physics - XII (ONLINE)** — India
  International Centre for Theoretical Science (**ICTS**), — 28/06/2021-09/07/2021

- **International school on: Gapless Fermions- from Fermi liquids to strange metals**, — Germany
  Max Planck Institute for the Physics of Complex Systems **(MPIPKS)**. — 17/02/2020-28/02/2020 .
  Poster Presentation: *Low-energy level spacing and entropy change across SYK non-Fermi liquid to a Fermi Liquid transition for finite-N*

National Conference, School and Presentation

- **QMAT conference** — India
  IIT Kanpur, — 19/09/2022-22/09/2022
  Talk: *Renyi entanglement entropy in Hubbard model within dynamical mean field theory*

- **APS-satellite meeting** — India
  International Centre for Theoretical Science (**ICTS**) — 15/03/2022-18/03/2022
  Talk: *Renyi entanglement entropy in Hubbard model within dynamical mean field theory*

- **Young Investigator Meet on Quantum Condensed Matter Theory** (**YIMQCMT**) — India
  ONLINE — 16/12/2020-18/12/2020
  Talk: *Quantum to classical crossover in many-body chaos in the quantum spherical p-spin glass model*

- **QMAT conference** — India
  Indian Institute of Science (**IISc**) — 08/07/2019-10/07/2019
  Poster Presentation: *Exact diagonalization study of low-energy spectra and thermalization across a non- Fermi liquid to Fermi liquid transition*

Teaching Assistance

- **Advanced Statistical Mechanics** — India
  Indian Institute of Science (**IISc**) — Aug/2021 - Nov/2021
  Course Instructor: Prof. Sumilan Banerjee

- **Advanced Condensed Matter** — India
  Indian Institute of Science (**IISc**) — Jan/2021 - Apr/2021
  Course Instructor: Prof. Chandan Dasgupta

- **Advanced Mathematical Methods** — India

Indian Institute of Science (**IISc**) Aug/2019 - Nov/2019
Course Instructor: Prof. Sumilan Banerjee

- **Thermal and Statistical Physics** India
  Indian Institute of Science (**IISc**) Jan/2019 - Apr/2019
  Course Instructor: Prof. Prerna Sharma

Prizes and Awards

- **Best Teaching Assistance (TA) Award** India
  Physical Science Department, IISc 2021

- **Prof. M A Viswamitra Award** India
  Top scorer in the first year of M.S. Integrated PhD programme, IISc 2017

- **INSPIRE Scholarship** India
  Recipient of this prestigious scholarship 2013 - 2016

Mentoring Activities

- **Indian Institute of Science (IISc)**, Mentoring of a Master student: Santanu Singh India
  Final Project PH250B for his M.Sc degree under Prof. Sumilan Banerjee Aug/2023-Mar/2024
  Thesis title: *Entanglement entropy of strange metal with spatially random interactions*

- **Indian Institute of Science (IISc)**, Mentoring of a Bachelor student: Saswata Mandal India
  Final project for his B.Sc degree under Prof. Sumilan Banerjee Aug/2021-Apr/2022
  Thesis title: *Entanglement Entropy in of Low dimensional Spin Systems*